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1 INTRODUCTION

One of the first articles [4] ... found that an independent Poisson distribution gives a reasonably accurate description of football scores. It achieved further improvements by applying a bivariate Poisson distribution ... The Negative Binomial distribution ... provides a better fit than the Poisson distribution [1], [2], [3] ... They found that the distributions do not follow the thin-tailed Poisson, nor the negative binomial distributions beyond the low scores, but heavy-tailed extremal distributions provide a better fit. In particular, the best fit of total scores was found to have a tail distribution with Gumbel α of 1, home and away scores were found to have a tail distribution with Fréchet α of 1.04 and 1.1, respectively. [49]

... they suggested a modified Poisson distribution [5] that accounts for the fact that the home and away scores are not independent. In particular, the proposed distribution function of the full time home and away scores is:

$$f_T(n_1, n_2) = \tau(n_1, n_2) * e^{-\Lambda_1} * \frac{\Lambda_1^{n_1}}{n_1!} * e^{-\Lambda_2} * \frac{\Lambda_2^{n_2}}{n_2!}$$

where n_1 and n_2 are the full time scores of the home and away teams, Λ_1 and Λ_2 are the average number of goals scored by the home and away teams during the game and the function $\tau(n_1, n_2)$ is responsible for the dependency between the scores of the two teams:

$$\begin{aligned}
\tau(n_1, n_2) = & \\
& 1 - \Lambda_1 * \Lambda_2 * \rho, \text{ if } n_1 = n_2 = 0, \\
& 1 + \Lambda_1 * \rho, \text{ if } n_1 = 0, n_2 = 1, \\
& 1 + \Lambda_2 * \rho, \text{ if } n_1 = 1, n_2 = 0, \\
& 1 - \rho, \text{ if } n_1 = 1, n_2 = 1, \\
& 1, \text{ otherwise}
\end{aligned}$$

They found that goal scoring intensities depend on the game time with intensity increasing steadily as the game progresses. [23] They also found that scoring intensities also depend on the current score. Therefore they proposed a model in which the goals scored by the two teams follow two Poisson processes with intensities that are functions of the current number of goals and time:

$$dN_t^i = d\aleph_t^{\lambda_i(N_t^1, N_t^2, t)}, i \in 1, 2,$$

where i = 1 refers to the home and i = 2 refers to the away team, N_t^i denotes the number of goals scored by team i, $\aleph_t^\lambda$ denotes a Poisson process of intensity λ and $\lambda_i(N_t^1, N_t^2, t)$ denotes the intensity function of team i which is a function of the current number of goals (N_t^1, N_t^2) and time t ... find that the model can be applied both to explain the distribution of final scores and also to valuing in-play spread bets.

The standard homogeneous Poisson process [17] ... develop analytical valuation formulae for in-play spread bets on goals and also on corners. A hierarchy of models suitable for in-play football betting has been studied by [42], starting from a homogeneous Poisson process with constant intensity, then introducing time-dependent intensities, finally introducing dependency on the current number of goals. He derived a valuation formula for in-play football bets in the general case when the intensity function is homogeneous in time:

$$\lambda_i(n_1, n_2, t) = \rho_i(n_1, n_2) * \bar{\xi}(t), i \in 1, 2,$$

where $\rho_i(n_1, n_2)$ is the score-dependent component and $\bar{\xi}(t)$ is the time-dependent component.The introduction of stochastic intensities has been suggested by [42] where the goal scoring intensities of the two teams depend not only on time and current score, but also on an independent Brownian driving process. In particular, he studied the case of the Cox-Ingerson-Ross process:

$$dx_t = \alpha(\Theta - x_t)dt + \sigma\sqrt{x_t}dW_t$$

Market rationality has been tested by [41] on the point spread betting market ... They concluded that statistical tests cannot reject market rationality while economic tests do reject market rationality. This implies that the betting market is reasonably efficient. An efficiency test of point spread betting markets has been conducted by [39] ... They concluded that market inefficiencies are present, but they tend to dissipate over time.

2 THE CONSTANT INTENSITY MODEL

Let us consider a probability space (Ω, F) that carries two independent Poisson processes N_t^1, N_t^2 with respective intensities μ_1, μ_2 and the filtration $(F_t), t \in [0, T]$, generated by these processes. Let time t = 0 denote the beginning and t = T the end of the game. The Poisson processes represent the number of goals scored by the teams, the superscript 1 refers to the home and 2 refers to the away team. The probability measure P is the real-world or physical probability measure. I assume that there exists a liquid market where three assets can be traded continuously with no transaction costs or any restrictions on short selling or borrowing. The first asset B_t is a risk-free bond that bears no interests, an assumption that is motivated by the relatively short time frame of a football game. The second and third assets S_t^1 and S_t^2 are such that their values at the end of the game are equal to the number of goals scored by the home and away teams, respectively.

The model is defined by the following asset dynamics:

$$\begin{aligned}
&B_t = 1, \\
&S_t^1 = N_t^1 + \lambda_1(T - t), \\
&S_t^2 = N_t^2 + \lambda_2(T - t),
\end{aligned}$$

where λ_1 and λ_2 are known real constants.

The time-t value of a bet is equal to the risk-neutral expectation of its value at the end of the game, formally:

$$X_t = E_Q[X_T|F_t]$$

A European bet is a bet with a value depending only on the final number of goals N_T^1, N_T^2, that is one of the form:

$$X_T = \Pi(N_T^1, N_T^2),$$

where Π is a known scalar function N x $N \to R$.

A typical example is a bet that pays out 1 if the home team scores more goals than the away team (home wins) and pays nothing otherwise, that is $\Pi(N_T^1, N_T^2) = 1(N_T^1 > N_T^2)$ where the function $1(A)$ takes the value of 1 if A is true and zero otherwise.

Another example is a bet that pays out 1 if the total number of goals is strictly higher than 2 and pays nothing otherwise, that is $\Pi(N_T^1, N_T^2) = 1(N_T^1 + N_T^2 > 2)$.

PROPOSITION: The time-t value of a European bet is given by the explicit formula:

$$X_t = X_t(t, N_t^1, N_t^2) = X_t(t, N_t^1, N_t^2, \lambda_1, \lambda_2) =$$
$$\Sigma_{n_1=N_t^1}^{\infty}[\Sigma_{n_2=N_t^2}^{\infty}[\Pi(n_1, n_2) * P(n_1 - N_t^1, \lambda_1(T-t)) * P(n_2 - N_t^2, \lambda_2(T-t))]],$$

where $P(N, \Lambda)$ is the Poisson probability, that is $P(N, \Lambda) = \frac{e^{-\Lambda} * \Lambda^N}{N!} if N \geq 0, and P(N, \Lambda) = 0$ otherwise.

The greeks are the values of the following forward difference operators (δ_1, δ_2) and partial derivative operator applied to the bet value:

$$\delta_1 X_t(N_t^1, N_t^2) = X_t(N_t^1 + 1, N_t^2) - X_t(N_t^1, N_t^2),$$
$$\delta_2 X_t(N_t^1, N_t^2) = X_t(N_t^1, N_t^2 + 1) - X_t(N_t^1, N_t^2),$$
$$\partial_t X_t(N_t^1, N_t^2) = lim_{dt \to 0}[\frac{1}{dt} X_{t+dt}(N_t^1, N_t^2)] - X_t(N_t^1, N_t^2),$$

The forward difference operators δ_1, δ_2 play the role of Delta and the partial derivative operator ∂_t plays the role of Θ in the Black-Scholes framework.

Bet type	Payoff $\Pi\left(N_T^1, N_T^2\right)$	Bet value $X_t\left(N_t^1, N_t^2, \lambda_1, \lambda_2\right)$
Home win	$\mathbf{1}\left(N_T^1 > N_T^2\right)$	$\sum_{k_1>k_2} \prod_{i=1}^2 P\left(k_i - N_t^i, \Lambda_i\right)$
Away win	$\mathbf{1}\left(N_T^1 < N_T^2\right)$	$\sum_{k_1<k_2} \prod_{i=1}^2 P\left(k_i - N_t^i, \Lambda_i\right)$
Draw	$\mathbf{1}\left(N_T^1 = N_T^2\right)$	$\sum_{k_1=k_2} \prod_{i=1}^2 P\left(k_i - N_t^i, \Lambda_i\right)$
Correct Score	$\mathbf{1}\left(N_T^1 = K_1, N_T^2 = K_2\right)$	$\prod_{i=1}^2 P\left(K_i - N_t^i, \Lambda_i\right)$
Over K	$\mathbf{1}\left(N_T^1 + N_T^2 > K\right)$	$\sum_{k=K+1}^{\infty} P\left(k - N_t^1 - N_t^2, \left(\Lambda_1 + \Lambda_2\right)\right)$
Under K	$\mathbf{1}\left(N_T^1 + N_T^2 < K\right)$	$\sum_{k=0}^{K-1} P\left(k - N_t^1 - N_t^2, \left(\Lambda_1 + \Lambda_2\right)\right)$
Odd	$\mathbf{1}\left(N_T^1 + N_T^2 = 1 \mod 2\right)$	$\exp\left[-\left(\Lambda_1 + \Lambda_2\right)\right] \cosh\left[\left(\Lambda_1 + \Lambda_2\right)\right]$
Even	$\mathbf{1}\left(N_T^1 + N_T^2 = 0 \mod 2\right)$	$\exp\left[-\left(\Lambda_1 + \Lambda_2\right)\right] \sinh\left[\left(\Lambda_1 + \Lambda_2\right)\right]$
Winning Margin	$\mathbf{1}\left(N_T^1 - N_T^2 = K\right)$	$\exp\left[-\left(\Lambda_1 + \Lambda_2\right)\right] \left(\frac{\Lambda_1}{\Lambda_2}\right)^{\frac{K-N_t^1+N_t^2}{2}} \cdot B_{\lvert K-N_t^1+N_t^2\rvert}\left(2\sqrt{\Lambda_1\Lambda_2}\right)$

The Next Goal Home bet [(i=1) or Away bet (i=2)] wins if the home team scores the next goal. The value of this bet is:

$$X_t = \frac{\lambda_i}{\lambda_1 + \lambda_2} * 1 - e^{-(\lambda_1+\lambda_2)(T-t)}.$$

Half Time / Full Time bets win if the half time and the full time is won by the predicted team or is a draw. Given that there are 3 outcomes in each halves, there are 9 bets in this category. For example, the value of the Half Time Home / Full Time Draw bet before the end of the first half is equal to:

$$X_t = \Sigma_{k_1>k_2}[\Sigma_{l_1=l_2}[P(k_1-N_t^1, \lambda_1(T_{\frac{1}{2}}-t))*P(k_2-N_t^2, \lambda_2(T_{\frac{1}{2}}-t))*P(l_1-k_1, \lambda_1(T-T_{\frac{1}{2}}))*P(l_2-k_2, \lambda_2(T-T_{\frac{1}{2}}))]].$$

In the second half, this bet either becomes worthless if the first half was not won by the home team or otherwise becomes equal to the Draw bet.

... how to calibrate the model on historical market prices. I apply a least squares approach in which I consider market prices of a set of bets and find model intensities that deliver model prices for these bets that are as close as possible to the market prices. Specifically, I minimize the sum of the square of the weighted differences between the model and market mid prices as a function of model intensities, using market bid-ask spreads as weights. The reason for choosing a bid-ask spread weighting is that I would like to take into account bets with a lower bid-ask spread with a higher weight because the price of these bets is more certain. Formally, the following expression is minimised:

$$R(\lambda_t^1, \lambda_t^2) = \sqrt{\frac{\frac{1}{n}\Sigma_{i=1}^n[X_t^{(i,MID)} - X_t^i(\lambda_t^1, \lambda_t^2, N_t^1, N_t^2)]}{(\frac{1}{2}X_t^{(i,SELL)} - X_t^{(i,BUY)})^2}},$$

where n is the total number of bets used, $X_t^{(i,BUY)}$ and $X_t^{(i,SELL)}$ are the best market buy and sell quotes of the i^{th} type of bet at time t, $X_t^{(i,MID)}$ is the market mid price which is the average of the best buy and sell quotes ... assumed the logarithm of the sum of the intensities of the two teams to follow a Wiener process of constant drift and volatility, that is:

$$dln(\lambda_t^1 + \lambda_t^2) = \mu dt + \sigma dW_t.$$

... I demonstrate market completeness and show that any bet can be replicated by dynamically trading in two linearly independent bets ... any European bet X_t can be replicated by dynamically trading in two linearly independent instruments Z_t^1 and Z_t^2:

$$X_t = X_0 + \int_0^t \psi_s^1 dZ_s^1 + \int_0^t \psi_s^2 dZ_s^2,$$

where the portfolio weights ψ_t^1, ψ_t^2 are equal to the solution of the equation:

$$\begin{pmatrix} \delta_1 Z_t^1 & \delta_1 Z_t^2 \\ \delta_2 Z_t^1 & \delta_2 Z_t^2 \end{pmatrix} \begin{pmatrix} \psi_t^1 \\ \psi_t^2 \end{pmatrix} = \begin{pmatrix} \delta_1 X_t \\ \delta_2 X_t \end{pmatrix}$$

where the values of the finite difference operators δ can be computed using the calibrated model intensities and the Kolmogorov forward equation where the value of European bet satisfies:

$$\partial_t X_t(N_t^1, N_t^2) = -\lambda_1 \delta_1 X_t(t, N_t^1, N_t^2) - \lambda_2 \delta_2 X_t(t, N_t^1, N_t^2).$$

It must be noted that if the scores become uneven and the game is close to finishing, the leading team is almost certain to win the game. In this case the values of these bets become insensitive to additional goals, because another goal does not significantly change the probabilities of winning. In this case the finite differences $\delta_i Z_t^j, i, j \in 1, 2$, become small. If the contract X_t that is being replicated is still sensitive to additional goals, then the portfolio weights ψ_t^i can become very high which leads to instabilities in the replication. This is similar to the problem of hedging a digital option with the underlying stock when the stock price is close to strike and the option is close to maturity. In order to avoid this problem, I perform replication only in those cases when the absolute number of any of the replicating instruments we need to hold in the portfolio is less than 50 ... hedging errors can sometimes be significant due to the fact the implied intensities are in fact not constant. In order to account for this effect, two potential extensions of the model could be considered: local intensity models where intensities are deterministic functions of the state and stochastic intensity models where intensities change randomly.

3 THE LOCAL INTENSITY MODEL

... an extension to the homogeneous Poisson model by allowing for state and time dependent intensities. The necessity for the model comes from the fact that neither the historical realised goal frequencies, nor the market implied intensities are constant, the distribution of goals doesn't follow a homogeneous Poisson distribution but is more heavy-tailed ... very similar to option markets where the Black-Scholes implied volatility is not constant, but depends on the option strike and therefore the distribution assumed by the market is more fat-tailed than a log-normal distribution. This becomes a practical problem when we are attempting to price an over-the-counter option or bet which is not traded on the market because it's unclear which implied volatility or intensity shall be used. the model needs to be extended in a way such that it is consistent with all market observable prices, that is it is able to reprice the intensity or volatility smile and produces the more heavy-tailed distributions. The answer in finance was Dupire's local volatility model ([31] and [45]) ... instead of assuming a constant volatility, the local volatility model assumes that the volatility is a function of spot and time, this function is referred to as the local volatility surface. Using the calibrated local volatility surface in a (PDE) or Monte-Carlo simulation exactly reprices all European options ... the implied intensity smile also exist in football betting, both in historical and market implied goal frequencies, that is the distribution of actual scores is more heavy-tailed than the standard Poisson distribution (historical frequencies) and that the market is aware of this fact (market implied intensities). Then I introduce the local intensity model where the goal intensity is not constant, but depends on the number of goals and time. I derive the Dupire's calibration formula for football betting that delivers a local intensity surface from either bet prices or implied intensities ... can be calibrated to more heavy-tailed distributions than the standard Poisson distribution. I remain in 1 dimension, that is I consider one team only for simplicity, but extension to two teams should be straightforward.

Within the Local Intensity model, the number of goals follows the following SDE:

$$dN_t = dN_t^{\lambda_{Loc}(N_t,t)},$$

where N_t denotes the total number of goals scored by the two teams up to time t, $N_t^{\lambda_{Loc}(N_t,t)}$ is a Poisson process of intensity $\lambda_{Loc}(N_t, t)$ and $\lambda_{Loc}(N_t, t)$ is the local intensity which is a function of the number of goals N_t and time t.

(Local intensity) function is defined by:

$$\lambda_{Loc}(K,T) = lim_{dt \to 0} \frac{1}{dt} P[N_{t+dt} = N_t + 1 | t = T, N_t = K].$$

The time evolution of the marginal distribution is described by the Kolmogorov forward equation:

$$\frac{\partial}{\partial T} f(K,T) = -\lambda_{Loc}(K,T) f(K,T) + \lambda_{Loc}(K-1,T) f(K-1,T),$$

where $f(K,T) = P[N_t = K, t = T]$ is the marginal density function.

... digital options with a payoff of $\Theta(S_T - K)$ (where $\Theta(x)$ is the Heaviside unit step function (0 if negative and 1 if positive)) can also be used in theory, Black-Scholes volatilities can also be implied from these products which are in general different from European implied volatilities and similar formulae can also be developed ... digital bets with a payoff of $\Theta(N_T - K)$ have a liquid market and are called Over-Under bets, where "over" refers to digital calls and "under" refers to digital puts. Arrow-Debreu-like bets with a payoff of $\delta^K_{N_T}$ (where δ^j_i denotes the Kronecker delta) are also traded in practice and are called Correct Score bets.

3.1 Over-Under Bets

The most liquidly traded bets on the total score are called Over-Under bets. The payoff is equal to 1 (that is, the bet wins) if $N_T \leq K$ and 0 otherwise, where K is called the strike of the bet. The value of an Over-Under bet of strike K and maturity T is denoted by $F(K,T)$ and is equal to the marginal cumulative distribution function of N_T:

$$F(K,T) = P[N_T \leq K].$$

The implied intensity of this bet is equal to the intensity of a Poisson distribution that has a cumulative distribution equal to the value of the bet. (Implied intensity of an Over-Under) bet of strike K and maturity T is defined by $\Lambda^{OU}_{Impl}(K,T)$ that satisfies:

$$F(K,T) = e^{-\Lambda^{OU}_{Impl}(K,T)} * \Sigma^K_{k=0} [\frac{(\Lambda^{OU}_{Impl}(K,T))^k}{k!}] = \frac{\Gamma(K+1, \Lambda^{OU}_{Impl}(K,T))}{K!},$$

where $\Gamma(n,x)$ denotes the incomplete gamma function. The local intensity surface can be calibrated from prices of Over-Under bets using the following formula:

$$\lambda_{Loc}(K,T) = -\frac{\frac{\partial}{\partial T}F(K,T)}{F(K,T)-F(K-1,T)}$$

The local intensity surface can be calibrated from the implied intensity surface using the following formula:

$$\lambda_{Loc}(K,T) = -\frac{e^{-\Lambda^{OU}_{Impl}(K,T)}(\Lambda^{OU}_{Impl})^K(K,T)\frac{\partial\Lambda^{OU}_{Impl}(K,T)}{\partial T}}{\Gamma(K+1,\Lambda^{OU}_{Impl}(K,T))-K\Gamma(K,\Lambda^{OU}_{Impl}(K-1,T)},$$

A first order expansion of the calibration formula if implied intensity changes slowly with strike can be obtained as follows:

$$\lambda_{Loc}(K,T) = \frac{\partial\Lambda^{OU}_{Impl}(K,T)}{\partial T}[1+K*\frac{\Lambda^{OU}_{Impl}(K-1,T)-\Lambda^{OU}_{Impl}(K,T)}{\Lambda^{OU}_{Impl}(K,T)}]^{-1}.$$

3.2 Correct Score Bets

Correct Score bets are also known as Arrow-Debreu securities and are also traded on the market, however the liquidity is somewhat less than the liquidity of Over-Under bets. The value of a Correct Score bet is equal to the probability $P[N_T = K]$. (Implied intensity of a Correct Score bet) of strike K and maturity T is $\Lambda^{CS}_{Impl}(K,T)$ that satisfies:

$$f(K,T) = e^{-\Lambda^{CS}_{Impl}(K,T)}\frac{(\Lambda^{CS}_{Impl}(K,T))^K}{K!}.$$

The local intensity surface can be calibrated from prices of Correct Score bets using the following formula:

$$\lambda_{Loc}(K,T) = -\frac{\frac{\partial}{\partial T}\Sigma^K_{k=0}f(k,T)}{f(K,T)}$$

The Local Intensity surface can be expressed as a function of Correct Score Implied Intensity surface as:

$$\lambda_{Loc}(K,T) = e^{\Lambda_I mpl^{CS}(K,T)}\frac{K!}{(\Lambda^{CS}_{Impl}(K,T))^K}\Sigma^K_{k=0}[e^{-\Lambda^{CS}_{Impl}(k,T)}\frac{(\Lambda^{CS}_{Impl}(k,T))^k}{k!}(1-\frac{k}{\Lambda^{CS}_{Impl}(k,T)})\frac{\partial\Lambda^{CS}_{Impl}(k,T)}{\partial T}].$$

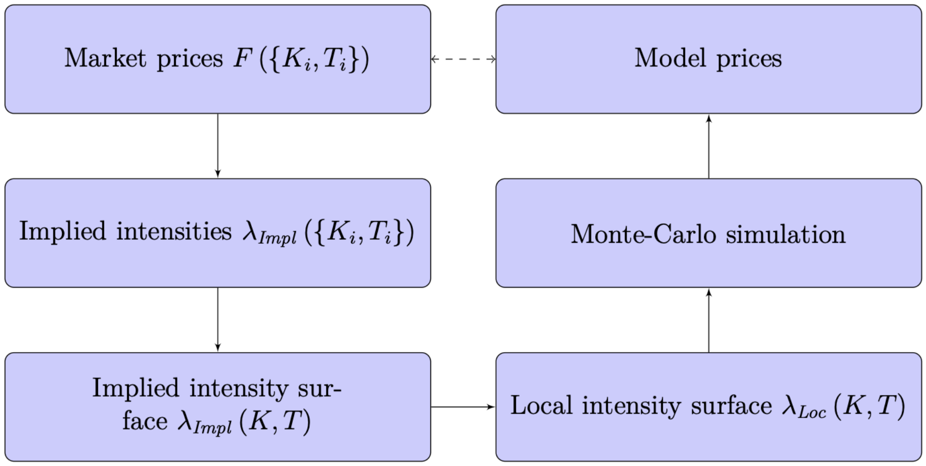

Figure 4.6: Summary of the steps involved in calibrating the Local Intensity model. Market prices of Over-Under bets $F(\{K_i, T_i\})$ are first transformed to implied intensities $\lambda_{Impl}(\{K_i, T_i\})$ which are then interpolated into an implied intensity surface $\lambda_{Impl}(K,T)$. Local intensity surface $\lambda_{Loc}(K,T)$ is constructed using the calibration formula 4.8 which in turn can be used to run a Monte-Carlo simulation to price an arbitrary bet. The prices computed in this way are going to be consistent with the original prices of Over-Under bets.